

```

Last login: Tue Jun 30 14:03:40 on ttys000
adityadave@Adityas-MacBook-Air-3 ~ % psql -d tpch-200mb
psql (18.4)
Type "help" for help.

tpch-200mb=# -- STEP 1: Cleanup
DROP VIEW IF EXISTS lineitem_view CASCADE;
RESET ROLE;

-- STEP 2: Plain view, no security_barrier
CREATE VIEW lineitem_view AS
SELECT l.*
FROM lineitem l
WHERE EXISTS (
    SELECT 1 FROM orders o
    WHERE o.o_orderkey = l.l_orderkey
    AND o.o_orderpriority IN ('1-URGENT', '2-HIGH')
);

-- STEP 3: Grants
GRANT SELECT ON lineitem_view TO alice;
GRANT SELECT ON orders TO alice;

-- STEP 4: Benchmark
SET ROLE alice;

EXPLAIN (ANALYZE, VERBOSE, BUFFERS)
SELECT sum(l_extendedprice * l_discount) AS revenue
FROM lineitem_view
WHERE l_quantity::text ~ '[1-9]$';
DROP VIEW
RESET
CREATE VIEW
GRANT
GRANT
SET
WARNING: Log 1: Planner Input:
DETAIL:  QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks false
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBENTRY
    :alias <>
    :eref
      {ALIAS
        :aliasname lineitem_view
        :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesstatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
      }
    :rtekind 1
  }
  :subquery
    {QUERY
      :commandType 1
      :querySource 0
      :canSetTag true
      :utilityStmt <>
      :resultRelation 0
      :hasAggs false
      :hasWindowFuncs false
      :hasTargetSRFs false
      :hasSubLinks true
      :hasDistinctOn false
      :hasRecursive false
      :hasModifyingCTE false
      :hasForUpdate false
      :hasRowSecurity false
      :hasGroupRTE false
      :isReturn false
      :cteList <>
      :rtable (
        {RANGETBENTRY
          :alias
            {ALIAS
              :aliasname l
              :colnames <>
            }
          :eref
            {ALIAS
              :aliasname l
              :colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesstatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
            }
          :rtekind 0
          :relid 17440
          :inh true
          :relkind r
          :rellockmode 1
          :perminfoindex 1
          :tablesample <>
          :lateral false
          :inFromCl true
          :securityQuals <>
        }
      )
      :rteperminfos (
        {RTEPERMISSIONINFO
          :relid 17440
          :inh true
          :requiredPerms 2
          :checkAsUser 10
          :selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
          :insertedCols (b)
          :updatedCols (b)
        }
      )
    }
  }
  :jointree
    {FROMEXPR
      :fromlist (
        {RANGETBLREF
          :rtindex 1
        }
      )
    }
  }
  :quals
    {SUBLINK
      :subLinkType 0
      :subLinkId 0
      :testexpr <>
      :opName <>
      :subselect
        {QUERY
          :commandType 1
          :querySource 0
          :canSetTag true
          :utilityStmt <>
          :resultRelation 0
          :hasAggs false
          :hasWindowFuncs false
          :hasTargetSRFs false
          :hasSubLinks false
        }
      }
    }
  }

```

```

:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:cteList <>
:rtable (
  {RANGETBLENTY
    :alias
      {ALIAS
        :aliasname o
        :colnames <>
      }
    :eref
      {ALIAS
        :aliasname o
        :colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_
          _totalprice" "o_orderdate" "o_orderpriority" "o_clerk"
            "o_shippriority" "o_comment")
      }
    :rtekind 0
    :relid 17428
    :inh true
    :relkind r
    :rellockmode 1
    :perminfoindex 1
    :tablesample <>
    :lateral false
    :inFromCl true
    :securityQuals <>
  }
)
:rteperminfos (
  {RTEPERMISSIONINFO
    :relid 17428
    :inh true
    :requiredPerms 2
    :checkAsUser 10
    :selectedCols (b 8 13)
    :insertedCols (b)
    :updatedCols (b)
  }
)
:jointree
{FROMEXPR
  :fromlist (
    {RANGETBLREF
      :rtindex 1
    }
  )
  :quals
    {BOOLEXP
      :boolop and
      :args (
        {OPEXP
          :opno 96
          :opfuncid 65
          :opresulttype 16
          :opretset false
          :opcollid 0
          :inputcollid 0
          :args (
            {VAR
              :varno 1
              :varattno 1
              :vartype 23
              :vartypmod -1
              :varcollid 0
              :varnullingrels (b)
              :varlevelsup 0
              :varreturningtype 0
              :varnosyn 1
              :varattnosyn 1
              :location -1
            }
            {VAR
              :varno 1
              :varattno 1
              :vartype 23
              :vartypmod -1
              :varcollid 0
              :varnullingrels (b)
              :varlevelsup 1
              :varreturningtype 0
              :varnosyn 1
              :varattnosyn 1
              :location -1
            }
          )
        }
        :location -1
      )
    }
    {SCALARARRAYOPEXP
      :opno 1054
      :opfuncid 1048
      :hashfuncid 0
      :negfuncid 0
      :useOr true
      :inputcollid 100
      :args (
        {VAR
          :varno 1
          :varattno 6
          :vartype 1042
          :vartypmod 19
          :varcollid 100
          :varnullingrels (b)
          :varlevelsup 0
          :varreturningtype 0
          :varnosyn 1
          :varattnosyn 6
          :location -1
        }
        {ARRAYEXP
          :array_typeid 1014
          :array_collid 100
          :element_typeid 1042
          :elements (
            {CONST
              :consttype 1042
              :consttypmod -1
              :constcollid 100
              :constlen -1
              :constbyval false
              :constisnull false
              :location -1
              :constvalue 12 [ 48 0 0 0 49 45 85 82 71 69 7
                0 84 ]
            }
            {CONST
              :consttype 1042
              :consttypmod -1
              :constcollid 100
              :constlen -1
              :constbyval false
              :constisnull false
              :location -1
              :constvalue 10 [ 40 0 0 0 50 45 72 73 71 72 ]
            }
          )
        }
        :multidims false
        :list_start -1
        :list_end -1
        :location -1
      )
    }
  )
  :location -1
}

```

```

    }
    }
    :location -1
  }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {CONST
      :consttype 23
      :consttypmod -1
      :constcollid 0
      :constlen 4
      :constbyval true
      :constisnull false
      :location -1
      :constvalue 4 [ 1 0 0 0 0 0 0 0 ]
    }
    :resno 1
    :resname ?column?
    :ressortgroupref 0
    :resorigtbl 0
    :resorigcol 0
    :resjunk false
  }
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:location -1
}
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 1
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 1
      :location -1
    }
    :resno 1
    :resname l_orderkey
    :ressortgroupref 0
    :resorigtbl 17440
    :resorigcol 1
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 2
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 2
      :location -1
    }
    :resno 2
    :resname l_partkey
    :ressortgroupref 0
    :resorigtbl 17440
    :resorigcol 2
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 3
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 3
      :location -1
    }
    :resno 3
    :resname l_suppkey
    :ressortgroupref 0
    :resorigtbl 17440
    :resorigcol 3
    :resjunk false
  }
  {TARGETENTRY
    :expr
    {VAR
      :varno 1
      :varattno 4
      :vartype 23
      :vartypmod -1
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 4
      :location -1
    }
    :resno 4
    :resname l_linenummer
    :ressortgroupref 0
    :resorigtbl 17440
    :resorigcol 4
    :resjunk false
  }
)
{TARGETENTRY
:expr

```

```

{VAR
:varno 1
:varattno 5
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 5
:location -1
}
:resno 5
:resname l_quantity
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 5
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 6
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 6
:location -1
}
}
:resno 6
:resname l_extendedprice
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 6
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 7
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 7
:location -1
}
}
:resno 7
:resname l_discount
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 7
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 8
:vartype 1700
:vartypmod 983046
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 8
:location -1
}
}
:resno 8
:resname l_tax
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 8
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 9
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 9
:location -1
}
}
:resno 9
:resname l_returnflag
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 9
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 10
:vartype 1042
:vartypmod 5
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 10
:location -1
}
}
:resno 10
:resname l_linestatus
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 10
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 11
:vartype 1002
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 11
:location -1
}
}
:resno 11
:resname l_shipdate
:ressortgroupref 0

```

```

:resorigtbl 17440
:resorigcol 11
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 12
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 12
:location -1
}
:resno 12
:resname l_commitdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 12
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 13
:vartype 1082
:vartypmod -1
:varcollid 0
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 13
:location -1
}
:resno 13
:resname l_receiptdate
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 13
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 14
:vartype 1042
:vartypmod 29
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 14
:location -1
}
:resno 14
:resname l_shipinstruct
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 14
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 15
:vartype 1042
:vartypmod 14
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 15
:location -1
}
:resno 15
:resname l_shipmode
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 15
:resjunk false
}
{TARGETENTRY
:expr
{VAR
:varno 1
:varattno 16
:vartype 1043
:vartypmod 48
:varcollid 100
:varnullingrels (b)
:varlevelsup 0
:varreturningtype 0
:varnosyn 1
:varattnosyn 16
:location -1
}
:resno 16
:resname l_comment
:ressortgroupref 0
:resorigtbl 17440
:resorigcol 16
:resjunk false
}
}
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location -1
:stmt_len -1
}
:security_barrier false
:relid 18520
:inh false
:relkind v
:relockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
)

```

```

:ртеperinfos (
  {RTEPERMISSIONINFO
  :reid 18520
  :inh true
  :requiredPerms 2
  :checkAsUser 0
  :selectedCols (b 12 13 14)
  :insertedCols (b)
  :updatedCols (b)
  }
)
:jointree
{FROMEXPR
:fromlist (
  {RANGETBLREF
  :rtindex 1
  }
)
:quals
  {OEXPR
  :opno 641
  :opfuncid 1254
  :opresulttype 16
  :opretset false
  :opcollid 0
  :inputcollid 100
  :args (
    {COERCEVIAIO
    :arg
    {VAR
    :varno 1
    :varattno 5
    :vartype 1700
    :vartypmod 983046
    :varcollid 0
    :varnullingrels (b)
    :varlevelsup 0
    :varreturningtype 0
    :varnosyn 1
    :varattnosyn 5
    :location 113
    }
    :resulttype 25
    :resultcollid 100
    :coerceformat 1
    :location 123
    }
    {CONST
    :consttype 25
    :consttypmod -1
    :constcollid 100
    :constlen -1
    :constbyval false
    :constisnull false
    :location 132
    :constvalue 11 [ 44 0 0 0 94 91 49 45 57 93 36 ]
    }
  )
  :location 130
  }
}
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
  {TARGETENTRY
  :expr
  {AGGREF
  :aggfnoid 2114
  :aggtype 1700
  :aggcollid 0
  :inputcollid 0
  :aggtranstype 0
  :aggargtypes (o 1700)
  :aggdirectargs <>
  :args (
    {TARGETENTRY
    :expr
    {OEXPR
    :opno 1760
    :opfuncid 1726
    :opresulttype 1700
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
      {VAR
      :varno 1
      :varattno 6
      :vartype 1700
      :vartypmod 983046
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 6
      :location 47
      }
      {VAR
      :varno 1
      :varattno 7
      :vartype 1700
      :vartypmod 983046
      :varcollid 0
      :varnullingrels (b)
      :varlevelsup 0
      :varreturningtype 0
      :varnosyn 1
      :varattnosyn 7
      :location 65
      }
    )
    :location 63
    }
  :resno 1
  :resname <>
  :ressortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
  }
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar false
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggssplit 0
:aggno -1
:aggtransno -1
:location 43
}
:resno 1
:resname revenue
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>

```

```

:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDeps <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: Log 2 :: After pull up At starting of Path Generation:
DETAIL: {QUERY
:commandType 1
:querySource 0
:canSetTag true
:utilityStmt <>
:resultRelation 0
:hasAggs true
:hasWindowFuncs false
:hasTargetSRFs false
:hasSubLinks true
:hasDistinctOn false
:hasRecursive false
:hasModifyingCTE false
:hasForUpdate false
:hasRowSecurity false
:hasGroupRTE false
:isReturn false
:ctelist <>
:rtable (
{RANGETBENTRY
:alias <>
:eref
{ALIAS
:aliasname lineitem_view
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 1
:subquery <>
:security_barrier false
:relid 18520
:inh false
:relkind v
:rellockmode 1
:perminfoindex 1
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname 1
:colnames <>
}
:eref
{ALIAS
:aliasname 1
:colnames ("l_orderkey" "l_partkey" "l_suppkey" "l_linenumber" "l_quantity" "l_extendedprice" "l_discount" "l_tax" "l_returnflag" "l_linesatus" "l_shipdate" "l_commitdate" "l_receiptdate" "l_shipinstruct" "l_shipmode" "l_comment")
}
:rtekind 0
:relid 17440
:inh false
:relkind r
:rellockmode 1
:perminfoindex 2
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
{RANGETBENTRY
:alias
{ALIAS
:aliasname o
:colnames <>
}
:eref
{ALIAS
:aliasname o
:colnames ("o_orderkey" "o_custkey" "o_orderstatus" "o_totalprice" "o_orderdate" "o_orderpriority" "o_clerk" "o_shippriority" "o_comment")
}
:rtekind 0
:relid 17428
:inh false
:relkind r
:rellockmode 1
:perminfoindex 3
:tablesample <>
:lateral false
:inFromCl true
:securityQuals <>
}
}
)
:rteperminfos (
{RTEPERMISSIONINFO
:relid 18520
:inh true
:requiredPerms 2
:checkAsUser 0
:selectedCols (b 12 13 14)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17440
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23)
:insertedCols (b)
:updatedCols (b)
}
{RTEPERMISSIONINFO
:relid 17428
:inh true
:requiredPerms 2
:checkAsUser 10
:selectedCols (b 8 13)
:insertedCols (b)
:updatedCols (b)
}
)
}
:jointree
{FROMEXPR
:fromlist (
{JOINEXPR
:jointype 4
:isNatural false
:larg
{FROMEXPR
:fromlist (

```

```

        {RANGETBLREF
        :rtindex 2
        }
    }
    :quals <>
    }
:arg
{RANGETBLREF
:rtindex 3
}
:usingClause <>
:join_using_alias <>
:quals (
    {OPEXPR
    :opno 96
    :opfuncid 65
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 0
    :args (
        {VAR
        :varno 3
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 1
        :location -1
        }
        {VAR
        :varno 2
        :varattno 1
        :vartype 23
        :vartypmod -1
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 1
        :location -1
        }
    )
    :location -1
    }
    {SCALARARRAYOPEXPR
    :opno 1054
    :opfuncid 1048
    :hashfuncid 0
    :negfuncid 0
    :useOr true
    :inputcollid 100
    :args (
        {VAR
        :varno 3
        :varattno 6
        :vartype 1042
        :vartypmod 19
        :varcollid 100
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 3
        :varattnosyn 6
        :location -1
        }
        {CONST
        :consttype 1014
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location -1
        :constvalue 48 [ -64 0 0 0 1 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0
        0 0 48 0 0 0 49 45 85 82 71 69 78 84 40 0 0 0 50 45 72 73 71 7
        2 0 0 ]
        }
    )
    :location -1
    }
    :alias <>
    :rtindex 0
    }
)
:quals (
    {OPEXPR
    :opno 641
    :opfuncid 1254
    :opresulttype 16
    :opretset false
    :opcollid 0
    :inputcollid 100
    :args (
        {COERCEVIAIO
        :arg
        {VAR
        :varno 2
        :varattno 5
        :vartype 1700
        :vartypmod 983046
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 5
        :location -1
        }
        :resulttype 25
        :resultcollid 100
        :coerceformat 1
        :location 123
        }
        {CONST
        :consttype 25
        :consttypmod -1
        :constcollid 100
        :constlen -1
        :constbyval false
        :constisnull false
        :location 132
        :constvalue 11 [ 44 0 0 0 94 91 49 45 57 93 36 ]
        }
    )
    :location 130
    }
    }
)
:mergeActionList <>
:mergeTargetRelation 0
:mergeJoinCondition <>
:targetList (
    {TARGETENTRY
    :expr
    {AGGREF
    :aggfnoid 2114
    :aggtype 1700
    :aggcollid 0
    :inputcollid 0
    :aggtranstype 0
    :aggargtypes (o 1700)

```

```
:aggdirectargs <>
:args (
  {TARGETENTRY
    :expr
      {OEXPR
        :opno 1760
        :opfuncid 1726
        :opresulttype 1700
        :opretset false
        :opcollid 0
        :inputcollid 0
      }
    :args (
      {VAR
        :varno 2
        :varattno 6
        :vartype 1700
        :vartypmod 983046
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 6
        :location -1
      }
      {VAR
        :varno 2
        :varattno 7
        :vartype 1700
        :vartypmod 983046
        :varcollid 0
        :varnullingrels (b)
        :varlevelsup 0
        :varreturningtype 0
        :varnosyn 2
        :varattnosyn 7
        :location -1
      }
    )
    :location 63
  }
  :resno 1
  :resname revenue
  :ressortgroupref 0
  :resorigtbl 0
  :resorigcol 0
  :resjunk false
}
)
:aggorder <>
:aggdistinct <>
:aggfilter <>
:aggstar false
:aggvariadic false
:aggkind n
:aggpresorted false
:agglevelsup 0
:aggsplit 0
:aggno -1
:aggtransno -1
:location 43
}
:resno 1
:resname revenue
:ressortgroupref 0
:resorigtbl 0
:resorigcol 0
:resjunk false
}
)
:override 0
:onConflict <>
:returningOldAlias <>
:returningNewAlias <>
:returningList <>
:groupClause <>
:groupDistinct false
:groupingSets <>
:havingQual <>
>windowClause <>
:distinctClause <>
:sortClause <>
:limitOffset <>
:limitCount <>
:limitOption 0
:rowMarks <>
:setOperations <>
:constraintDecls <>
:withCheckOptions <>
:stmt_location 0
:stmt_len 0
}

WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 641 :opfuncid 1254 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 100 :args ({COERCEVIAIO :arg {VAR :varno 2 :varattno 5 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 5 :location -1} :resulttype 25 :resultcollid 100 :coerceformat 1 :location -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 11 [ 44 0 0 0 94 91 49 45 57 93 36 ]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b 2) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1} {CONST :consttype 25 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 11 [ 44 0 0 0 94 91 49 45 57 93 36 ]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 2) :required_relds (b 2) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b 2) :orclause <> :rinfo_serial 3 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1} {VAR :varno 2 :varattno 7 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 7 :location -1} {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1}
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {SCALARARRAYOEXPR :opno 1054 :opfuncid 1048 :hashfuncid 0 :negfuncid 0 :useOr true :inputcollid 100 :args ({VAR :varno 3 :varattno 6 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 6 :location -1} {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 48 [ -64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 49 45 85 82 71 69 78 84 40 0 0 50 45 72 73 71 72 0 0 ]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :outer_relds (b) :left_relds (b) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 3 :varattno 6 :vartype 1042 :vartypmod 19 :varcollid 100 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 6 :location -1} {CONST :consttype 1014 :consttypmod -1 :constcollid 100 :constlen -1 :constbyval false :constisnull false :location -1 :constvalue 48 [ -64 0 0 0 1 0 0 0 0 0 0 0 18 4 0 0 2 0 0 0 1 0 0 0 48 0 0 49 45 85 82 71 69 78 84 40 0 0 50 45 72 73 71 72 0 0 ]} :location -1} :is_pushed_down true :can_join false :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 0 :security_level 0 :num_base_rels 1 :clause_relds (b 3) :required_relds (b 3) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b 2) :orclause <> :rinfo_serial 2 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec -1 :mergeopfamilies <> :left_em <> :right_em <> :outer_is_left false :hashjoinoperator 0 :left_bucketsize -1 :right_bucketsize -1 :left_mcvfreq -1 :right_mcvfreq -1 :left_hasheqoperator 0 :right_hasheqoperator 0}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1}
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1}
WARNING: is_parallel_safe: INPUT qual = ({RESTRICTINFO :clause {OEXPR :opno 96 :opfuncid 65 :opresulttype 16 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} {VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1} :location -1} :is_pushed_down true :can_join true :pseudoconstant false :has_clone false :is_clone false :leakproof false :has_volatile 2 :security_level 0 :num_base_rels 2 :clause_relds (b 2 3) :required_relds (b 2 3) :incompatible_relds (b) :outer_relds (b) :left_relds (b 2) :right_relds (b 3) :orclause <> :rinfo_serial 4 :eval_cost.startup -1 :eval_cost.per_tuple 0 :norm_selec -1 :outer_selec 0.3991666666666667 :mergeopfamilies (o 1976) :left_em {EQUIVALENCMEMBER :em_expr {VAR :varno 2 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 2 3)}} :right_em {EQUIVALENCMEMBER :em_expr {VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1} :em_relds (b 2) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 2 3)}} :right_em {EQUIVALENCMEMBER :em_expr {VAR :varno 3 :varattno 1 :vartype 23 :vartypmod -1 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 3 :varattnosyn 1 :location -1} :em_relds (b 2 3) :em_is_const false :em_is_child false :em_datatype 23 :em_jdomain {JOINDOMAIN :jd_relds (b 2 3)}} :outer_is_left false :hashjoinoperator 96}) | parallel_safe = true | reason = PROPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1
```

```
on -1} {VAR :varno 2 :varattno 7 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 7 :location -1}}
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1} {VAR :varno 2 :varattno 7 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 7 :location -1}) | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({AGGREF :aggfnoid 2114 :aggtype 1700 :aggcollid 0 :inputcollid 0 :aggtranstype 2281 :aggargtypes (o 1700) :aggdirectargs <> :args ({TARGETENTRY :expr {OEXPR :opno 1760 :opfuncid 1726 :opresulttype 1700 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1} {VAR :varno 2 :varattno 7 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 7 :location -1}) :location -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false}) :aggorder <> :aggdistinct <> :aggfilter <> :aggstar false :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({AGGREF :aggfnoid 2114 :aggtype 1700 :aggcollid 0 :inputcollid 0 :aggtranstype 2281 :aggargtypes (o 1700) :aggdirectargs <> :args ({TARGETENTRY :expr {OEXPR :opno 1760 :opfuncid 1726 :opresulttype 1700 :opretset false :opcollid 0 :inputcollid 0 :args ({VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1} :location -1} :resno 1 :resname <> :ressortgroupref 0 :resorigtbl 0 :resorigcol 0 :resjunk false}) :aggorder <> :aggdistinct <> :aggfilter <> :aggstar false :aggvariadic false :aggkind n :aggpresorted false :agglevelsup 0 :aggsplit 0 :aggno 0 :aggtransno 0 :location -1}) | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = ({VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1} {VAR :varno 2 :varattno 7 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 7 :location -1})
WARNING: is_parallel_safe: OUTPUT qual = ({VAR :varno 2 :varattno 6 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 6 :location -1} {VAR :varno 2 :varattno 7 :vartype 1700 :vartypmod 983046 :varcollid 0 :varnullingrels (b) :varlevelsup 0 :varreturningtype 0 :varnosyn 2 :varattnosyn 7 :location -1}) | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPPARALLEL_SAFE
WARNING: is_parallel_safe: INPUT qual = <>
WARNING: is_parallel_safe: OUTPUT qual = <> | parallel_safe = true | reason = PROPPARALLEL_SAFE
```

QUERY PLAN

```
Finalize Aggregate (cost=39986.73..39986.74 rows=1 width=32) (actual time=187.159..188.153 rows=1.00 loops=1)
  Output: sum((l1_extendedprice * l1_discount))
  Buffers: shared hit=14355 read=13469
-> Gather (cost=39986.51..39986.72 rows=2 width=32) (actual time=186.603..188.141 rows=3.00 loops=1)
  Output: (PARTIAL sum((l1_extendedprice * l1_discount)))
  Workers Planned: 2
  Workers Launched: 2
  Buffers: shared hit=14355 read=13469
-> Partial Aggregate (cost=38986.51..38986.52 rows=1 width=32) (actual time=183.723..183.724 rows=1.00 loops=3)
  Output: PARTIAL sum((l1_extendedprice * l1_discount))
  Buffers: shared hit=14355 read=13469
  Worker 0: actual time=182.147..182.148 rows=1.00 loops=1
  Buffers: shared hit=4785 read=4433
  Worker 1: actual time=182.601..182.602 rows=1.00 loops=1
  Buffers: shared hit=4769 read=4466
-> Parallel Hash Semi Join (cost=7458.20..38936.61 rows=9979 width=12) (actual time=183.720..183.721 rows=0.00 loops=3)
  Output: l1_extendedprice, l1_discount
  Hash Cond: (l1_orderkey = o.o_orderkey)
  Buffers: shared hit=14355 read=13469
  Worker 0: actual time=182.144..182.145 rows=0.00 loops=1
  Buffers: shared hit=4785 read=4433
  Worker 1: actual time=182.598..182.599 rows=0.00 loops=1
  Buffers: shared hit=4749 read=4466
-> Parallel Seq Scan on public.lineitem l (cost=0.00..31301.77 rows=24999 width=16) (actual time=151.577..151.577 rows=0.00 loops=3)
  Output: l1_orderkey, l1_partkey, l1_suppkey, l1_linenum, l1_quantity, l1_extendedprice, l1_discount, l1_tax, l1_returnflag, l1_linestatus, l1_shipdate, l1_commi
tdate, l1_receiptdate, l1_shipinstruct, l1_shipmode, l1_comment
  Filter: ((l1_quantity)::text ~ '[1-9]$'::text)
  Rows Removed by Filter: 399990
  Buffers: shared hit=9754 read=12798
  Worker 0: actual time=151.576..151.576 rows=0.00 loops=1
  Buffers: shared hit=3263 read=4275
  Worker 1: actual time=151.587..151.587 rows=0.00 loops=1
  Buffers: shared hit=3241 read=4293
-> Parallel Hash (cost=6834.50..6834.50 rows=49896 width=4) (actual time=31.896..31.896 rows=40048.00 loops=3)
  Output: o.o_orderkey
  Buckets: 131072 Batches: 1 Memory Usage: 5760kB
  Buffers: shared hit=4601 read=671
  Worker 0: actual time=30.552..30.553 rows=38332.00 loops=1
  Buffers: shared hit=1522 read=158
  Worker 1: actual time=30.552..30.553 rows=38220.00 loops=1
  Buffers: shared hit=1508 read=173
-> Parallel Seq Scan on public.orders o (cost=0.00..6834.50 rows=49896 width=4) (actual time=0.088..0.261 rows=40048.00 loops=3)
  Output: o.o_orderkey
  Filter: (o.o_orderpriority = ANY ('{1-URGENT,2-HIGH}'))::bpchar[])
  Rows Removed by Filter: 59952
  Buffers: shared hit=4601 read=671
  Worker 0: actual time=0.018..0.251 rows=38332.00 loops=1
  Buffers: shared hit=1522 read=158
  Worker 1: actual time=0.010..0.251 rows=38220.00 loops=1
  Buffers: shared hit=1508 read=173

Planning:
  Buffers: shared hit=197
  Planning Time: 5.736 ms
  Execution Time: 188.465 ms
(53 rows)

tpch-200mb=>
```